Development of a Open Source Autopilot for QUT use

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Acknowledgement of Traditional Owners

The Queensland University of Technology (QUT) acknowledges the Turrbal and Yugara, as the First Nations owners of the lands where QUT now stands. We pay respect to their Elders, lores, customs and creation spirits. We recognise that these lands have always been places of teaching, research and learning.

QUT acknowledges the important role Aboriginal and Torres Strait Islander people play within the QUT community.

Project Overview

Task Description:
This project involves refactoring, building, programming and testing a Open Source flight module design to serve as a replacement for current hardware used in UAV units at QUT. The objective of this project is to create a In-house solution to flight controller hardware capable of replacing the difficult to source off the shelf version (Pixhawk 6).

The work presented here was derived off the v1 version produced by **KAI(Wesley) 2022**

The final demonstrator should include:

1. Loading Open source firmware with customisations
2. Connect to QGroundControl
3. Basic operations for manned flight on a simple airframe

Design License

AUAV Pixracer is an open hardware design that is a evolution of the PixHawk design, following the OSHW 1.1 definition licensed under the Creative Commons Attribution-ShareAlike 3.0 Unported (CC BY-SA 3.0) license.

Project Progress

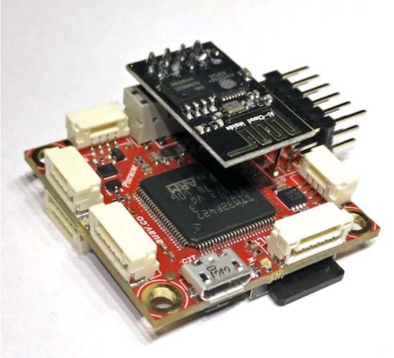
PixRacer

The Pixracer is the common name for the FMUv4 generation of Pixhawk autopilots. It has been designed primarily for the demanding requirements of small multicopters, but of course can be used on planes and other vehicles which require no more than 6 pwm outputs for controlling escs and motors. Pixracer is available from the mRobotics

The Pixhawk® XRacer board family is optimized for small racing quads and planes. In contrast to Pixfalcon and Pixhawk it has in-built WiFi, new sensors, convenient full servo headers, CAN and supports 2M flash.

Key Features

- Main System-on-Chip: STM32F427VIT6 rev.3
  - CPU: 180 MHz ARM Cortex® M4 with single-precision FPU
  - RAM: 256 KB SRAM (L1)
- JST GH connectors
- microSD (logging)
- Futaba S.BUS and S.BUS2 / Spektrum DSM2 and DSMX / Graupner SUMD / PPM input / Yuneec ST24
- FrSky® telemetry port
- OneShot PWM out (configurable)
- Optional: Safety switch and buzzer



DISPLAY : ALL CATEGORIES 47 > AIRCRAFT 1653 > DRONES 270 > FLIGHT CONTROLLERS & ACCESSORIES 28

Pixracer (Xracer) Autopilot V1.0 FMU V4 Generation

PRICE: A$91.51

Discontinued Product

Please continue searching for a similar product.

1 year warranty

SKU: 571000095-0

SHARE :

PIXRACER PRO Autopilot Xracer PX4...

$251.19 ~~$440~~

AliExpress

PIXRACER R15 Autopilot Xracer PX4 Pixhawk...

$175.39 ~~$264~~

AliExpress

Free delivery

Pixhawk PX4 Autopilot 2.4.8 32 Bit Flight...

$147.89 (INR 9,500)

aerokartindia.in

Pixracer R15 Flight Controller Designed F...

$192.75 (USD 136)

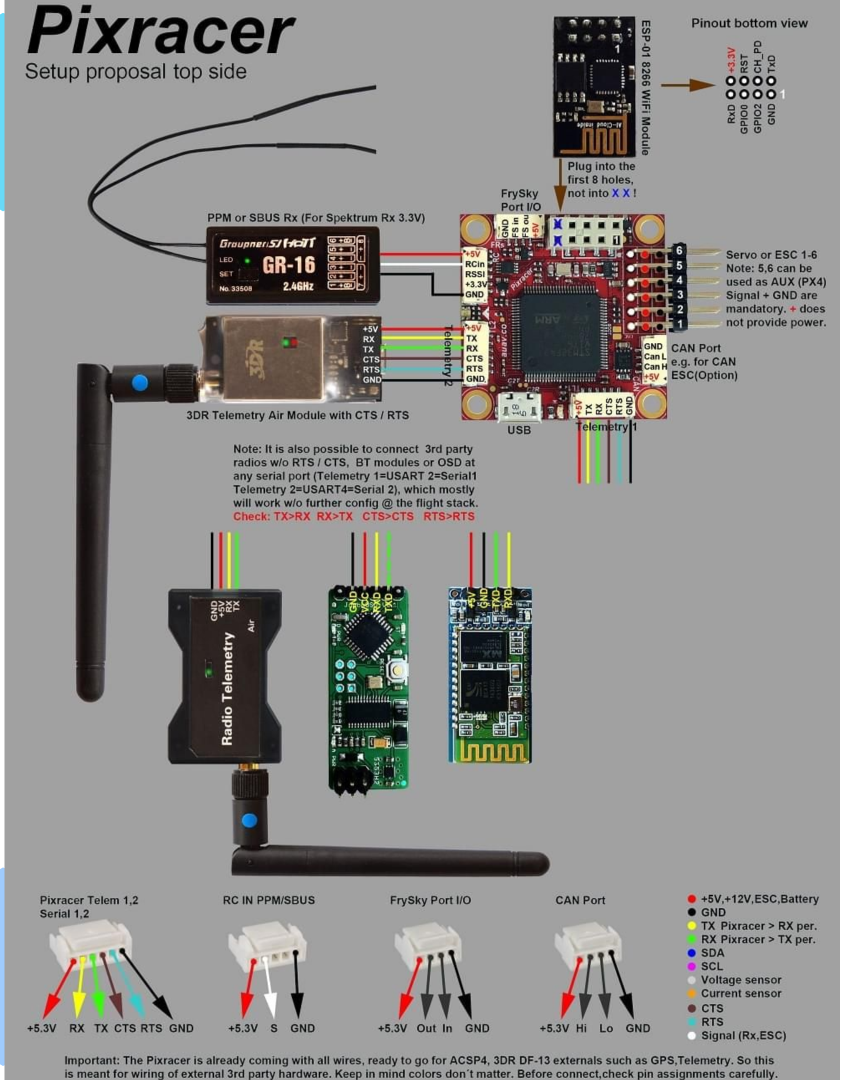
Alibaba.com

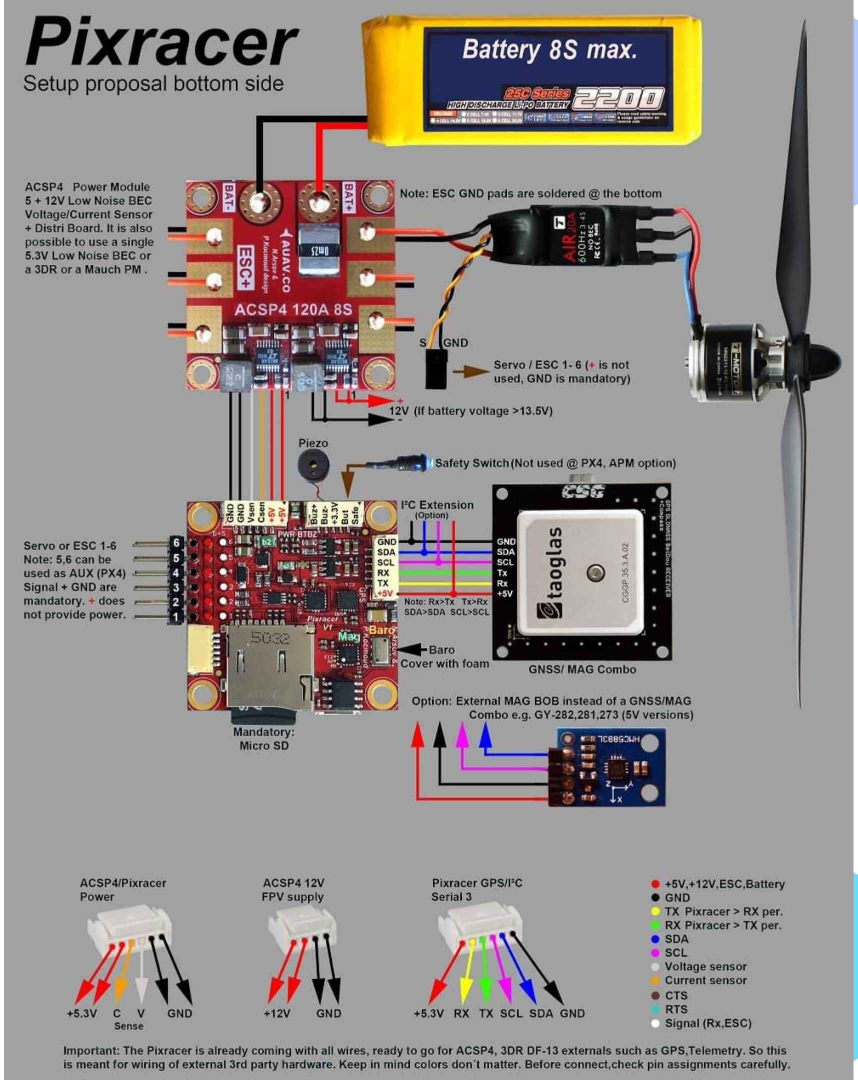
PIXRACER R15 Autopilot Xracer PX4 Pixhawk...

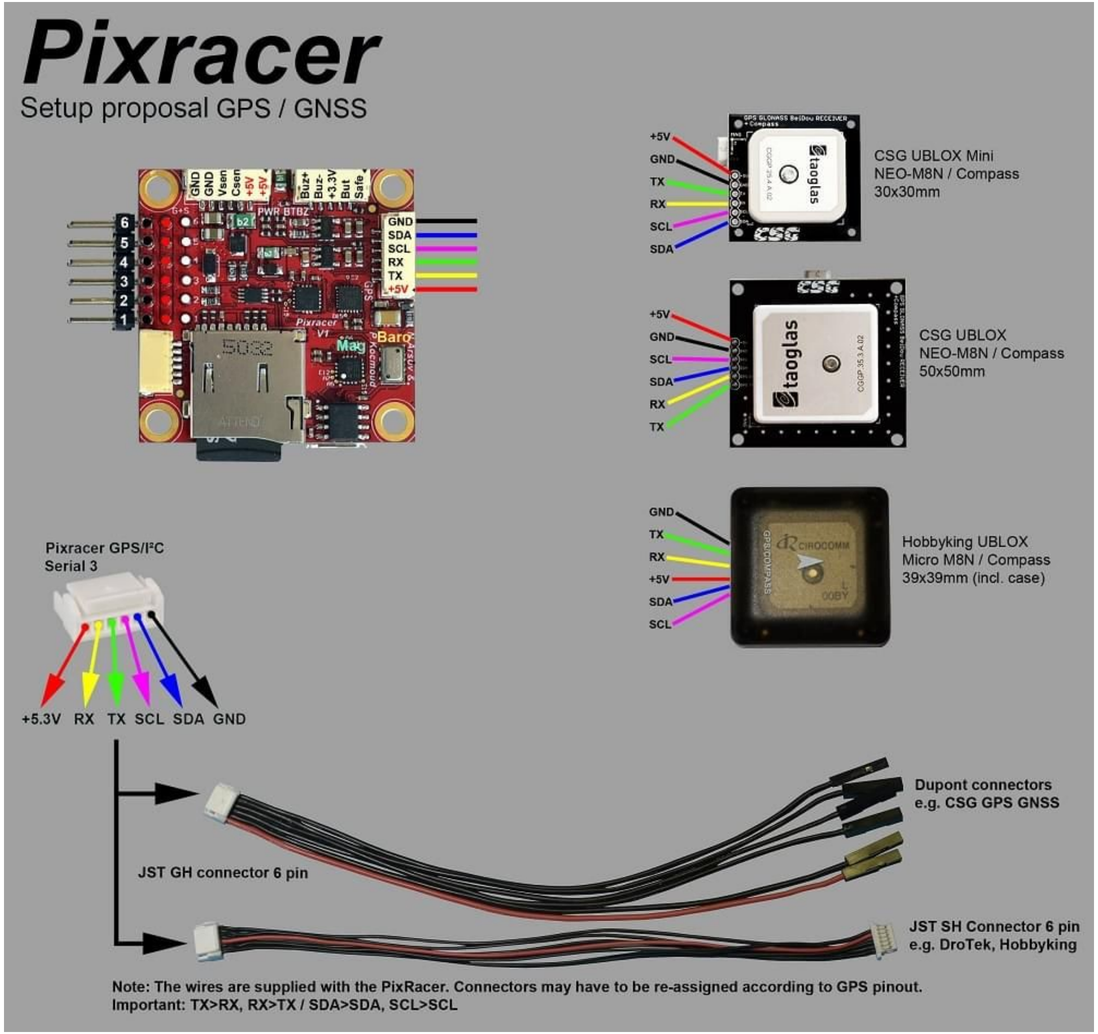
$104.69 ~~$248~~

AliExpress

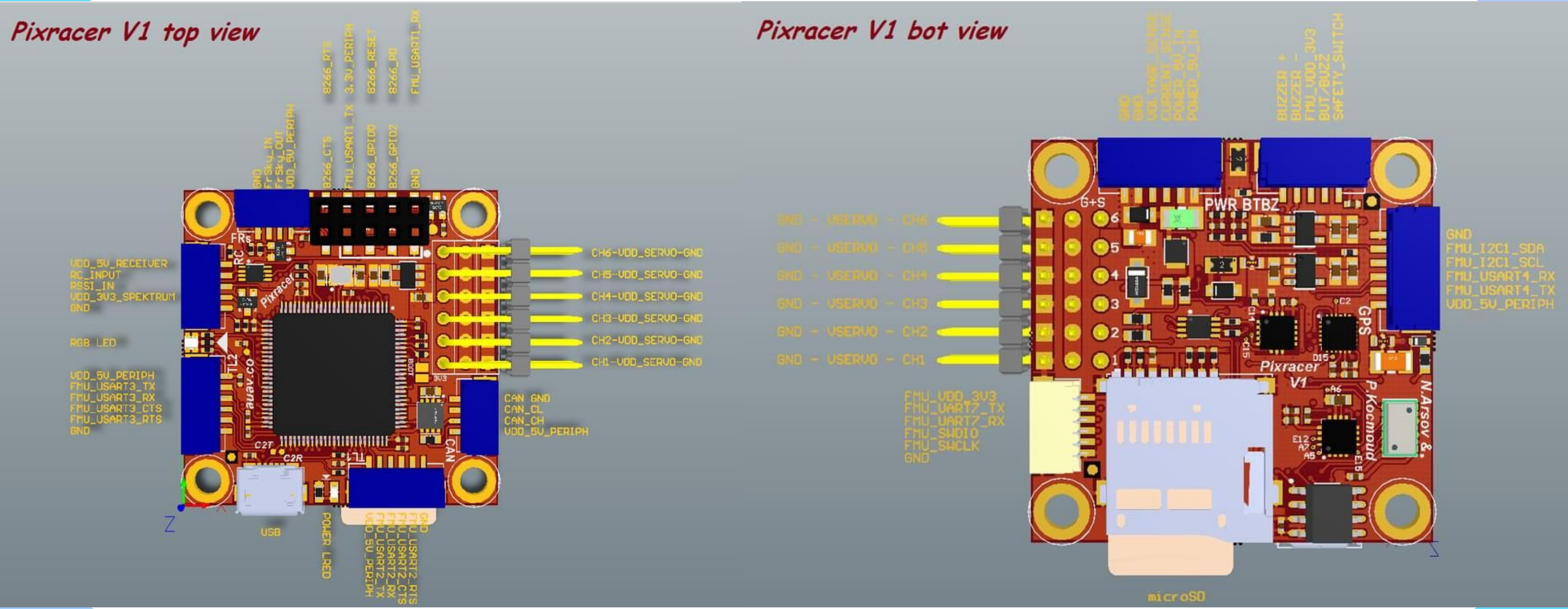
Free delivery


Pixracer
Setup proposal top side
Pinout bottom view
ESP-01 8266 WiFi Module
+3.3V RST CH_PD TxD
RxD GPIO0 GPIO2 GND
Plug into the first 8 holes, not into X X !
FrySky Port I/O
PPM or SBUS Rx (For Spektrum Rx 3.3V)
GR-16
2.4GHz
+5V RCin RSSI +3.3V GND
Servo or ESC 1-6
Note: 5,6 can be used as AUX (PX4)
Signal + GND are mandatory. + does not provide power.
+5V RX TX CTS RTS GND
+5V TX RX CTS RTS GND
Telemetry 2
GND Can L Can H +5V
CAN Port e.g. for CAN ESC(Option)
3DR Telemetry Air Module with CTS / RTS
USB
Telemetry 1
Note: It is also possible to connect 3rd party radios w/o RTS / CTS, BT modules or OSD at any serial port (Telemetry 1=USART 2=Serial1 Telemetry 2=USART4=Serial 2), which mostly will work w/o further config @ the flight stack.
Check: TX>RX RX>TX CTS>CTS RTS>RTS
Radio Telemetry
Pixracer Telem 1,2 Serial 1,2
+5.3V RX TX CTS RTS GND
RC IN PPM/SBUS
+5.3V S GND
FrySky Port I/O
+5.3V Out In GND
CAN Port
+5.3V Hi Lo GND
+5V,+12V,ESC,Battery
GND
TX Pixracer > RX per.
RX Pixracer > TX per.
SDA
SCL
Voltage sensor
Current sensor
CTS
RTS
Signal (Rx,ESC)
Important: The Pixracer is already coming with all wires, ready to go for ACSP4, 3DR DF-13 externals such as GPS,Telemetry. So this is meant for wiring of external 3rd party hardware. Keep in mind colors don´t matter. Before connect,check pin assignments carefully.


Pixracer
Setup proposal bottom side
Battery 8S max.
2200
ACSP4 Power Module
5 + 12V Low Noise BEC
Voltage/Current Sensor
+ Distri Board. It is also possible to use a single 5.3V Low Noise BEC or a 3DR or a Mauch PM .
Note: ESC GND pads are soldered @ the bottom
ACSP4 120A 8S
S GND
Servo / ESC 1- 6 (+ is not used, GND is mandatory)
12V (If battery voltage >13.5V)
Piezo
Safety Switch (Not used @ PX4, APM option)
I²C Extension (Option)
GND GND Vsen Csen +5V +5V Buz+ Buz- +3.3V But Safe
Servo or ESC 1-6
Note: 5,6 can be used as AUX (PX4)
Signal + GND are mandatory. + does not provide power.
GND SDA SCL RX TX +5V
Note: Rx>Tx Tx>Rx SDA>SDA SCL>SCL
Baro Cover with foam
GNSS/ MAG Combo
Mandatory: Micro SD
Option: External MAG BOB instead of a GNSS/MAG Combo e.g. GY-282,281,273 (5V versions)
ACSP4/Pixracer Power
+5.3V C V GND
Sense
ACSP4 12V FPV supply
+12V GND
Pixracer GPS/I²C Serial 3
+5.3V RX TX SCL SDA GND
+5V,+12V,ESC,Battery
GND
TX Pixracer > RX per.
RX Pixracer > TX per.
SDA
SCL
Voltage sensor
Current sensor
CTS
RTS
Signal (Rx,ESC)
Important: The Pixracer is already coming with all wires, ready to go for ACSP4, 3DR DF-13 externals such as GPS,Telemetry. So this is meant for wiring of external 3rd party hardware. Keep in mind colors don´t matter. Before connect,check pin assignments carefully.



Pixracer
Setup proposal GPS / GNSS
CSG UBLOX Mini
NEO-M8N / Compass
30x30mm
+5V
GND
TX
RX
SCL
SDA
CSG UBLOX
NEO-M8N / Compass
50x50mm
+5V
GND
SCL
SDA
RX
TX
Hobbyking UBLOX
Micro M8N / Compass
39x39mm (incl. case)
GND
TX
RX
+5V
SDA
SCL
GND
SDA
SCL
RX
TX
+5V
Pixracer GPS/I²C
Serial 3
+5.3V RX TX SCL SDA GND
Dupont connectors
e.g. CSG GPS GNSS
JST GH connector 6 pin
JST SH Connector 6 pin
e.g. DroTek, Hobbyking
Note: The wires are supplied with the PixRacer. Connectors may have to be re-assigned according to GPS pinout.
Important: TX>RX, RX>TX / SDA>SDA, SCL>SCL


Pixracer V1 top view
Pixracer V1 bot view
8266_RTS
3.3V_PERIPH
8266_RESET
8266_PD
FMU_USART1_RX
GND
FrSky_IN
FrSky_OUT
VDD_5V_PERIPH
8266_CTS
FMU_USART1_TX
8266_GPIO0
8266_GPIO2
GND
VDD_5V_RECEIVER
RC_INPUT
RSSI_IN
VDD_3V3_SPEKTRUM
GND
RGB LED
VDD_5V_PERIPH
FMU_USART3_TX
FMU_USART3_RX
FMU_USART3_CTS
FMU_USART3_RTS
GND
CH6-VDD_SERVO-GND
CH5-VDD_SERVO-GND
CH4-VDD_SERVO-GND
CH3-VDD_SERVO-GND
CH2-VDD_SERVO-GND
CH1-VDD_SERVO-GND
CAN_GND
CAN_CL
CAN_CH
VDD_5V_PERIPH
USB
POWER LRED
GND
FMU_USART2_RTS
FMU_USART2_CTS
FMU_USART2_RX
FMU_USART2_TX
VDD_5V_PERIPH
FRs
RC
Pixracer
auav.co
TL2
C2T
C2R
CAN
GND
GND
VOLTAGE_SENSE
CURRENT_SENSE
POWER_5V_IN
POWER_5V_IN
BUZZER +
BUZZER -
FMU_VDD_3V3
BUT/BUZZ
SAFETY_SWITCH
PWR BTBZ
G+S
GND - VSERVO - CH6
GND - VSERVO - CH5
GND - VSERVO - CH4
GND - VSERVO - CH3
GND - VSERVO - CH2
GND - VSERVO - CH1
GND
FMU_I2C1_SDA
FMU_I2C1_SCL
FMU_USART4_RX
FMU_USART4_TX
VDD_5V_PERIPH
GPS
FMU_VDD_3V3
FMU_UART7_TX
FMU_UART7_RX
FMU_SWDIO
FMU_SWCLK
GND
Pixracer V1
P.Kocmoud
N.Arsov &
microSD

Pixracer Specifications

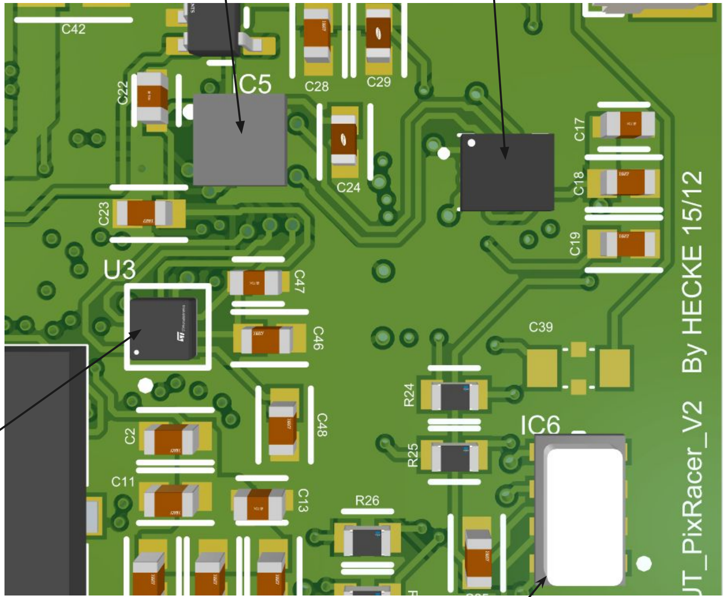
- Interfaces/Connectivity
  - Wifi: ESP-01 802.11bgn Flashed with MavESP8266
  - MicroSD card reader
  - Micro USB
  - RGB LED
  - GPS (serial + I2C)
  - TELEM1/TELEM2
  - Wifi serial
  - FrSky Telemetry serial(see note)
  - Debug connector (serial + SWD)
  - Connectors: GPS+I2C, RC-IN, PPM-IN, RSSI, SBus-IN, Spektrum-IN, USART3 (TxD, RxD, CTS, RTS), USART2 (TxD, RxD, CTS, RTS), FRSky-IN, FRSky-OUT, CAN, USART8 (TxD, RxD), ESP8266 (full set), SERVO1-SERVO6, USART7 (TxD, RxD), JTAG (SWDIO, SWCLK), POWER-BRICK (VDD, Voltage, Current, GND), BUZZER-LED_BUTTON.
- Dimensions
  - 36 x 36mm with 30.5 x 30.5mm hole grid with 3.2mm holes
- Processor:
  - MCU - STM32F427VIT6 rev.3
  - Ultra low noise LDOs for sensors and FMU
  - FRAM - FM25V02-G
- Sensors
  - Gyro/Accelerometer: Invensense MPU9250 Accel / Gyro / Mag (4 KHz)
  - Gyro/Accelerometer: Invensense ICM-20608 Accel / Gyro (4 KHz)
  - Barometer: MS5611
  - Compass: Honeywell HMC5983 magnetometer with temperature compensation
- Power
  - 5-5.5VDC from USB or PowerBrick connector. Optional/recommended ACSP4 +5V/+12V Power Supply.

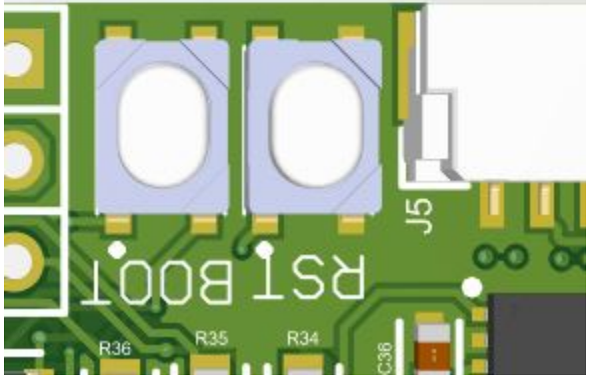
Modified Specifications

Different Sensors
-Gyro/Accelerometer: Invensense IMC42688 Accel / Gyro
-Gyro/Accelerometer: Invensense ICM-20608-G Accel / Gyro
Barometer: MS561101BA03-50 (SAME)
Compass: LIS3MDLTR Mag

Small Placement Changes (SD card, C and R Components. Tracers)(see v1 files)

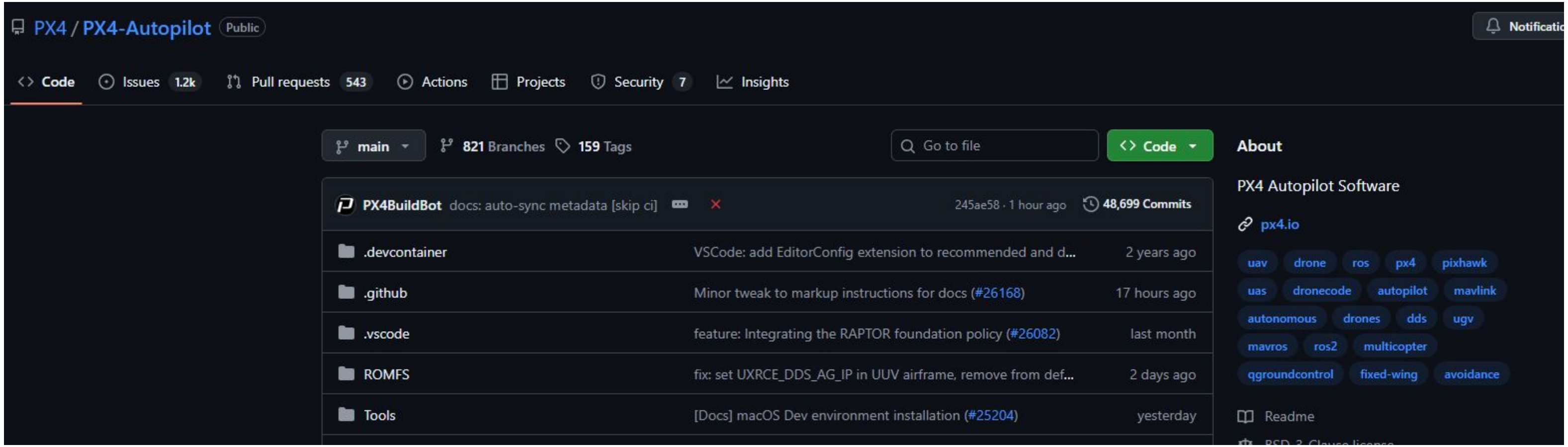
New Interfaces including Reset Button and BOOT0 Button





Firmware

The Px4 firmware loaded on the stm chip using DFU, Small modifications were required, including modifying drivers and the chip select pins for SPI lines. Mavlink 8266 firmware was also required for the external wifi esp32 ESP-01 chip that sits on top of the board for wifi telemetry. This was done with a Arch linux computer with small issues appearing, they were fixed. Custom firmware to test sensor communication was also designed, with the onboard led showing different colours for different communication failures.



Current Board Design Issues (moving to v3)

Issue 1: D4 label missing from board
Issue 2: Labels too small to read
Issue 3: No thermal relief on the ground pads
Issue 4: Board name out of date
Issue 5: USB data lines not same length
Issue 6: Rotate one of the accelerometer sensors 90 degrees so they are both not the same orientation
Issue 7: Add breakout for serial debugging output(or use DFU lines without breaking board)
Issue 8: Reverse SD card footprint for better access

Issue 9: STM chip not connected to 3.3v...
Issue 10: If Hand producing, place all small components on single side for reflow/solder mask use, then hand solder other side.
Issue 11: Review sensor interference due to tracers/other components etc
Issue 12: Barometer may be damaged during hot air soldering so ensure care is taken during hand soldering
Issue 13: Board could be bigger/ more holes for mounting/case mounting
Issue 14: Ethernet header to support non-wifi mode for pi usage??

Issue 15: Maybe consider replacing GH-JST connectors with more modern solution ??

Future work

-Flight test the prototype to test design validity (TODOTODAY)

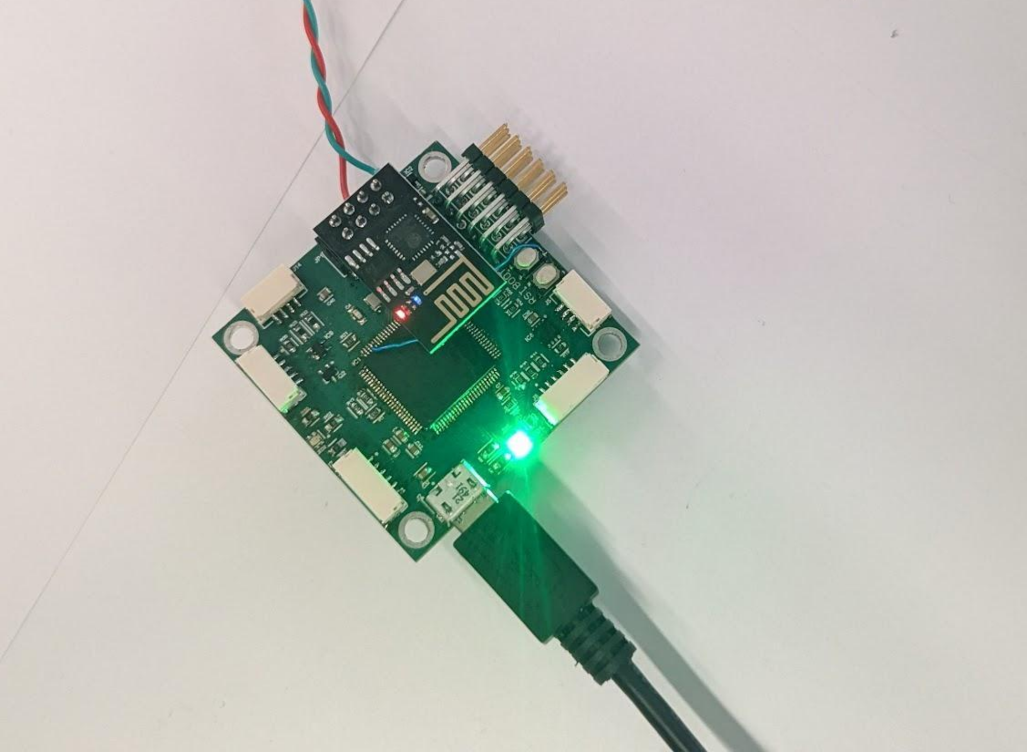
-Fixing Issues and setting up external production

-Design outer case with correct specs for the sensors, these include the barometer foam protection, chip vibration isolation techniques(rubber/foam spacer)

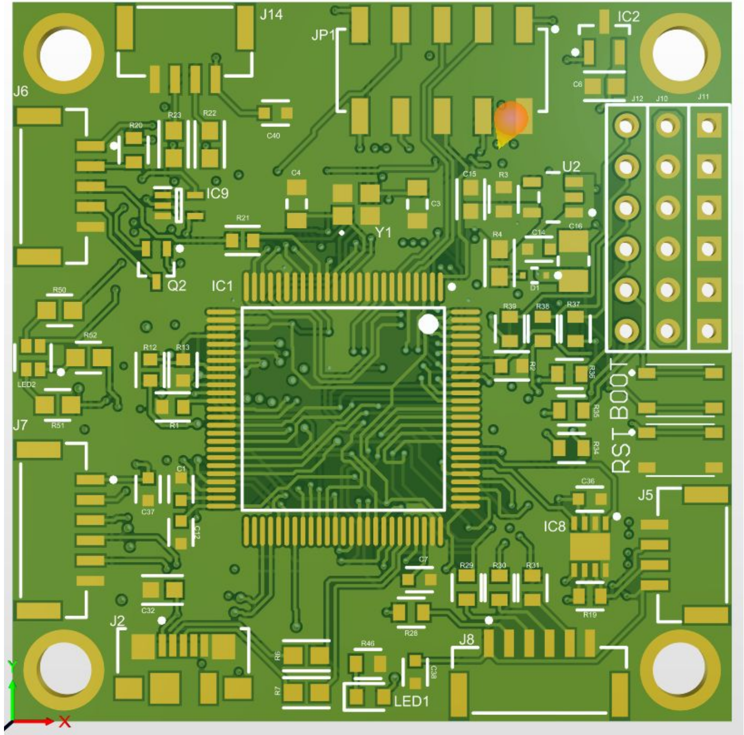
-Validate external component connections, PX4 firmware and QGroundcontrol configurations to ensure they are correct and accurate

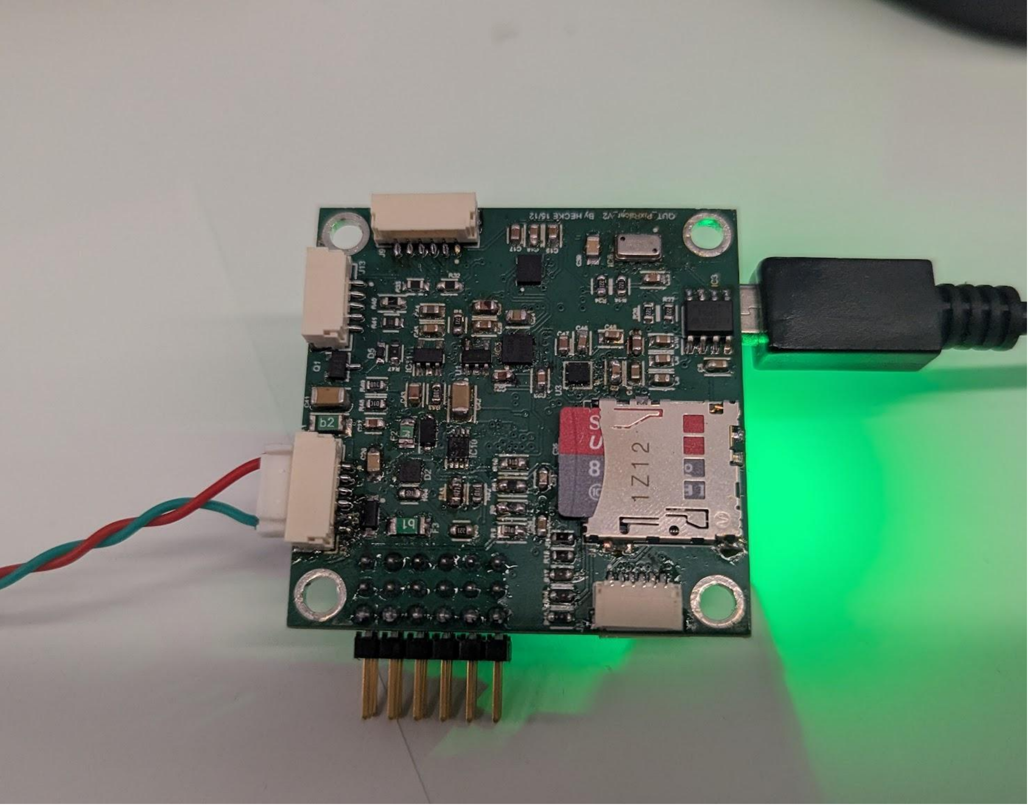
If the unit requires certain connections or electronics, modifying the design to allow for these electronics may be required (PI ethernet connection, Camera using Mavlink protocols, etc)

Consider turning this into a pi header with the required isolation for the whole system, this would be quite a change to the board but possible with the right CAD configs

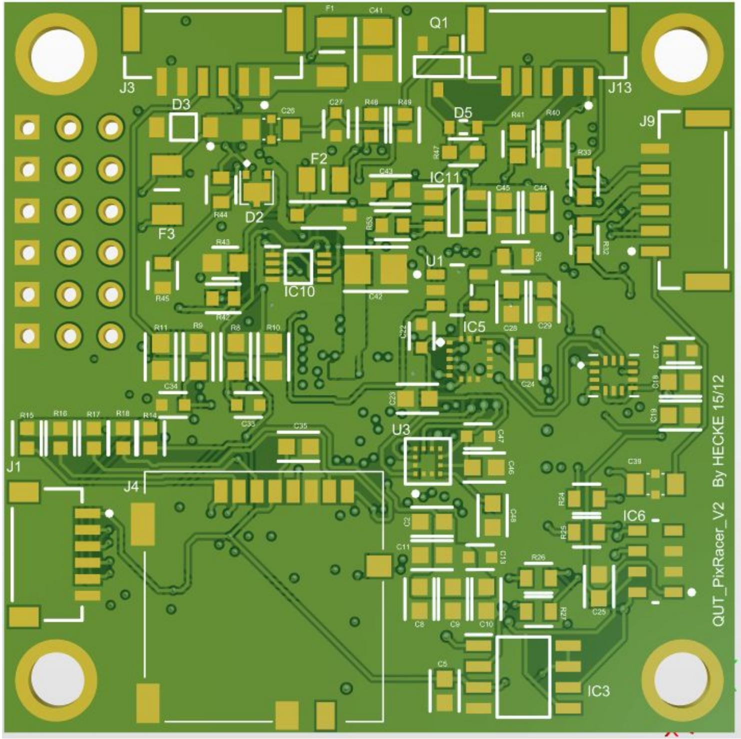


TOP





BOTTOM



Done